

Dimension	ELT	ETL
Where transform happens	Inside bigquery after loading	Before loading into BigQuery
Raw data preserved ?	Yes - full raw data preserved	No - data is transformed before loading so raw data is not preserved in BigQuery
Latency profile	Batch / near real time	Since, data is used for reporting purpose - near real time streaming not required in this case.
Transformation rigidity	Flexible	As no fixed reporting schema yet, repeated work would be required incase of any changes.
Compute location	GCP BigQuery	Outside BigQuery.
Cost driver	Higher as require larger storage space	Comparatively lower as data is already transformed.
Main risk if skipped	No latest data available for reporting	Sensitive data of conflict zones will be masked, and difference in version amount regions will lead to wrong column ingestion of data.

Considering the following factors, we can conclude that ETL would be a better choice for MSF - field operations data platform case. The conclusion is based on below facts

- 1) Patient data sensitivity - Conflict zones data is masked, incase of ELT we load both masked and unmasked data into same table and while unmasking the data we have to iterate too complete data. Where as incase of ETL we unmask the required zones before loading, reducing the operations performed.
- 2) Analytics requirements - In this case ELT is preferable as no fixed reporting schema can result in frequent analytical changes.
- 3) Data Format - Slight difference in versions such as column name, data formats, ETL is preferred as we can streamline data from all zones and load into one single table post transformation.